SRI SIDDHARTHA INSTITUTE OF TECHNOLOGY, TUMAKURU

(A constituent college of Sri Siddhartha Academy of Higher Education, Tumakuru)

22EC304: Analog Electronics Circuits

TEST-1 Time: 1.00Hr

Answer all the questions and assume the missing data:

		M	C	B
1	Design a full wave rectifier with 'C' filter with capacitor C= 1200 pF, frequency f= 50 Hz, and Ripple factor γ = 0.002.	04	1	3
2	Design and explain the clipping circuit for the following transfer characteristics.	06	1	2,3

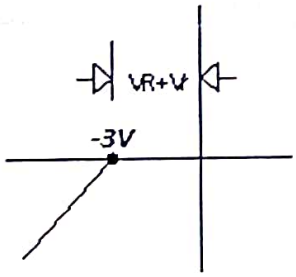

Fig. 2(a)

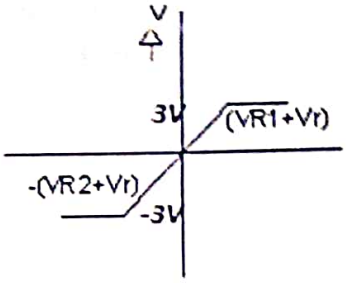

Fig. 2(b)

		M	C	B
3	Design and explain the following clamping circuit: i. To clamp positive peak at 2V ii. To clamp negative peak at $V\gamma$ = 0.7 V	06	1	2,3
4	Deduce the current expression for NMOS enhancement mode MOSFET operating in triode region.	08	2	2,3
5	With the help of small signal model, explain the T – Model of MOSFET.	06	2	2

NOTE: M is marks, C is CO and B is Blooms level

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22EC304/18EC304: ANALOG ELECTRONICS CIRCUITS

TEST2

Max Marks: 30

Q.No		Marks	CO	BL
1.	Explain the common source MOSFET amplifier with a ***source resistance***. Write neat circuit diagram and derive the expressions for voltage gain A_v, input R_i, overall voltage gain G_v and R_o	6	2	2
2.	Explain operation of MOS differential pair with a ***common-mode input voltage***.	6	2	2
3.	List the ***important properties of negative feedback*** and explain any two in detail.	8	2	2
4.	With a neat circuit diagram, explain the working of ***Colpitts oscillator.***	4	2	2
5.	Draw and explain the ***internal block diagram of an operational amplifier*** (μA741).	6	3	2

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22EC304: Analog Electronics Circuit

Max Marks:20 **TEST3** Time:1.00Hr.

Answer all the questions

		M	C	B
1	With the help of circuit and waveform explain the working of zero crossing detector.	5	4	2
2	Design a symmetric Schmitt trigger with UTP=3V and LTP = -3V. Assume Vsat = 12V.	5	4	3
3	With the help of circuit and waveform explain the working of non-inverting comparator.	5	4	2
4	Briefly explain the working of first order low pass filter	5	4	2

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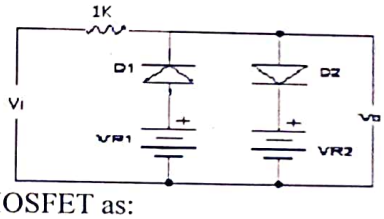
B.E., SEMESTER END EXAMINATION JAN - FEB 2024

22EC304: ANALOG ELECTRONICS CIRCUITS

TIME: 3.00 Hrs **SEM: III** **MAX MARKS: 100**

NOTE: Answer any five full questions selecting one full question from each choice.

		M	CO	Bl.
1.a)	With a neat circuit and waveform, derive an expression for ripple factor of full wave rectifier with capacitor filter (center tapped).	10	1	2
b)	Derive an equation between i_D - V_{DS} relationship for N-channel enhancement mode MOSFET. And also sketch the VI characteristic curve.	10	2	3
	OR			
2.a)	Draw the circuit diagram, waveform for positive and negative peak clampers and explain its operation.	8	1	2
b)	Analyze the clipper circuits and draw its output waveform and transfer curve.	4	1	4



		M	CO	Bl.
c)	Explain the operation of MOSFET as: i. Switch ii. Linear amplifier	8	2	2
3.a)	Draw T-equivalent circuit for small signal operation model of MOSFET.	6	2	1
b)	Explain the common source MOSFET amplifier with necessary equations and circuit diagram.	10	2	2
c)	Explain the MOS differential pair with a common-mode input voltage.	4	2	2
	OR			
4.a)	Explain the common source MOSFET amplifier with source resistance and derive the expressions for input R_i, gain A_v, overall voltage gain G_v and R_o.	10	2	2
b)	Explain common mode rejection ratio (CMRR) for both single ended output and differential output of a differential MOS and derive expressions for the same.	10	2	2
5.a)	Mention four basic feedback topologies and explain any two in details.	8	3	2
b)	Explain the following with respect to op-amp: i. Common mode rejection ratio ii. Power supply rejection ratio iii. Input and Output Offset voltages iv. Slew Rate	8	3	2
c)	Find the frequency of oscillation of Hartley oscillator with L1= 1mH, L2= 1.1mH & C=100μF.	4	3	3
	OR			

6.a)	List the important properties of negative feedback and explain any one in detail.	6	3	1
b)	With help of neat block diagram, explain the internal block diagram of operational amplifier.	6	3	
c)	Explain the Colpitts oscillator to obtain the sustained oscillation and also write the equation to find frequency of oscillation.	8	3	2
7.a)	Explain the working of comparator circuit using opamp with neat circuit diagram and waveform.	10	3	2
b)	Explain the working of Schmitt trigger circuit using positive feedback op-amp. And also write the waveform representing input and output.	10	3	2

OR

8.a)	Design an op-amp Schmitt trigger circuit with LTP=2V, UTP=6V and Vsat=±12V. Draw the input - output waveform for the same.	10	3	3
b)	With a neat circuit diagram and waveform, explain how a zero-crossing detector works where AC input is applied to inverting terminal.	10	4	2
9.a)	Explain the working of internal circuit diagram of 555 timer with help of neat circuit.	10	4	2
b)	Explain the working of RC-phase shift oscillator with neat circuit diagram. And also write the equation for frequency of oscillation.	10	4	2

OR

10.a)	With neat diagram and waveforms, explain the working of monostable multivibrator using 555 timer.	10	4	2
b)	Explain the working of class B push-pull amplifier with neat circuit diagram.	10	4	2

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